



Lab Report - 01

Experiment on 01

Course Name: Structured Programming Lab

Course Code: CSE-122

Submitted to

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Lab Report-01

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05	Write a C program to compute the perimeter and area of a circle with a given radius.
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07	Write a C program that accepts two items' weight and number of purchases (floating point values) and calculates their average value.

Signature :

Date : 24/06/2025

Problem-1: Write a C program to print your name, date of birth, and ID.

Expected Output:

Name:Alexandra Abramov

DOB : July 14, 1975

ID : 000-000-000

Source Code:

```
1  #include <stdio.h>
2  int main ()
3  {
4
5      printf("Name :Mahjabin Anika\n");
6      printf("DOB :January 26, 2004\n");
7      printf("ID :251-115-083\n");
8
9
10     return 0;
11 }
12
```

Output:

```
Name :Mahjabin Anika
DOB :January 26, 2004
ID :251-115-083

Process returned 0 (0x0)   execution time : 0.172 s
Press any key to continue.
|
```

Problem-2:Write a C program to print a block E using the hash (#), where the E has a height of seven characters and a width of six and five characters. And also print a very large 'C'.

Expected Output:

```
#####  
#  
#  
#####  
#  
#  
#####  
#####  
##  ##  
#  
#  
#  
#  
#  
##  ##  
#####
```

Source Code:

```
1  #include <stdio.h>
2  int main()
3  {
4      printf("#####\n");
5      printf("#\n");
6      printf("#\n");
7      printf("#####\n");
8      printf("#\n");
9      printf("#\n");
10     printf("#####\n\n\n");
11
12     printf(" #####\n");
13     printf("##  ##\n");
14     printf("#\n");
15     printf("#\n");
16     printf("#\n");
17     printf("#\n");
18     printf("#\n");
19     printf("##  ##\n");
20     printf(" #####\n");
21
22     return 0;
23
24 }
25
```

Output:

```
#####  
#  
#  
####  
#  
#  
#####
```

```
#####  
##  ##  
#  
#  
#  
#  
#  
##  ##  
#####
```

```
Process returned 0 (0x0)   execution time : 0.161 s  
Press any key to continue.
```

Problem-3: Write a C program to print the following characters in reverse.

Test Characters: 'X', 'M', 'L'

Expected Output:

The reverse of XML is LMX

Source Code:

```
1  #include<stdio.h>
2  int main()
3  {
4
5      char char1='X', char2='M', char3='L';
6      printf("The revers of %c%c%c is %c%c%c \n", char1, char2, char3, char3, char2, char1);
7
8      return 0;
9
10
11 }
12
```

Output:

```
The revers of XML is LMX
```

```
Process returned 0 (0x0)   execution time : 0.139 s
```

```
Press any key to continue.
```

Problem-4: Write a C program to compute the perimeter and area of a rectangle with a height of 7 inches and a width of 5 inches.

Expected Output:

Perimeter of the rectangle = 24 inches

Area of the rectangle = 35 square inches

Source Code:

```
1  #include<stdio.h>
2  int main()
3  {
4      int height = 7 , width = 5 ;
5      int perimeter = 2 * (height + width);
6      int area = height * width ;
7
8      printf(" Perimeter of the rectangle =%d inches \n",perimeter );
9      printf(" Area of the rectangle =%d square inches\n",area);
10
11     return 0;
12 }
13
```

Output:

```
Perimeter of the rectangle =24 inches
Area of the rectangle =35 square inches

Process returned 0 (0x0)   execution time : 0.147 s
Press any key to continue.
```


Problem-5: Write a C program to compute the perimeter and area of a circle with a given radius.

Sample Input:

Enter a radius:

6

Expected Output:

Perimeter of the Circle = 37.680000 inches

Area of the Circle = 113.040001 square inches

Source Code:

```
1  #include<stdio.h>
2  #define pi 3.14
3  int main ()
4  {
5      float radius ;
6      printf("Enter a radius:\n");
7      scanf("%f", &radius );
8
9      float perimeter = 2 * pi * radius;
10     float area = pi * radius * radius ;
11
12     printf("\nPerimeter of the circle = %f inches\n",perimeter);
13     printf("Area of the circle = %f square inches\n",area);
14
15     return 0;
16
17 }
18
```

Output:

```
Enter a radius:
6

Perimeter of the circle = 37.680000 inches
Area of the circle = 113.040001 square inches

Process returned 0 (0x0)   execution time : 3.375 s
Press any key to continue.
```

Problem-6: Write a C program to convert specified days into years, weeks and days. Note: Ignore leap year.

Sample Input:

Number of days: 1329

Expected Output :

Years: 3

Weeks: 33

Days: 3

Source Code:

```
1  #include<stdio.h>
2  int main()
3  {
4      int totaldays , years , weeks , days ;
5      printf("Number of days:");
6      scanf("%d",&totaldays);
7
8      years=totaldays/365;
9      weeks=(totaldays%365)/7;
10     days=(totaldays%365)%7;
11
12     printf("\nYears:%d\n",years);
13     printf("weeks:%d\n",weeks);
14     printf("Days:%d\n",days);
15     return 0;
16 }
17
```

Output:

```
Number of days:1329
```

```
Years:3
```

```
weeks:33
```

```
Days:3
```

```
Process returned 0 (0x0)   execution time : 4.550 s
Press any key to continue.
```

Problem-7: Write a C program that accepts two items' weight and number of purchases (floating point values) and calculates their average value.

Sample Input:

Weight - Item1: 15

No. of item1: 5

Weight - Item2: 25

No. of item2: 4

Expected Output:

Average Value = 19.444444

Source Code:

```
e x *Untitled1.c x
1  #include<stdio.h>
2  int main ()
3  {
4      float weight1, weight2, weight3, num1, num2, average;
5      printf ("Weight-Item1:");
6      scanf ("%f", &weight1);
7      printf ("No. of item1:");
8      scanf ("%f", &num1);
9
10     printf ("Weight-Item2:");
11     scanf ("%f", &weight2);
12     printf ("No. of item2:");
13     scanf ("%f", &num2);
14
15     average=(weight1*num1+weight2*num2) / (num1+num2);
16     printf ("\nAverage value=%f\n", average);
17
18     return 0;
19 }
```

Output:

```
Weight-Item1:15
No. of item1:5
Weight-Item2:25
No. of item2:4

Average value=19.444445

Process returned 0 (0x0)   execution time : 29.826 s
Press any key to continue.
|
```